

GIDDINGS LEE COUNTY AP, TX, USA

WMO: 722113

Lat: 30.169N

Lon: 96.980W

Elev: 485

StdP: 14.44

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53979

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.1	31.5	14.3	11.6	40.0	18.4	14.1	43.0	20.9	53.0	18.8	55.5	5.0	10	0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.6	99.9	73.8	98.2	73.6	96.1	73.9	79.4	88.6	78.3	87.6	77.5	86.8	6.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	145.3	81.7	76.4	140.2	81.0	75.2	134.7	80.3	43.2	88.6	42.1	87.8	41.2	86.9	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.4	15.6	12.9	DB	22.6	102.7	4.9	2.3	19.0	104.3	16.1	105.7	13.3	107.0	9.7	108.6
			WB	21.6	79.2	4.8	1.8	18.2	80.5	15.4	81.5	12.7	82.5	9.2	83.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	68.9	51.1	54.4	61.8	68.2	75.3	82.2	84.2	85.4	79.7	70.4	60.3	52.6
	DBStd	14.20	9.66	10.29	8.62	7.27	5.81	3.49	3.01	3.56	4.95	7.95	9.41	9.51
	HDD50	289	102	66	14	2	0	0	0	0	0	1	21	83
	HDD65	1615	441	316	164	50	5	0	0	0	0	42	198	399
	CDD50	7179	137	190	379	547	784	967	1061	1097	891	634	329	163
	CDD65	3032	11	20	64	145	325	517	596	632	442	210	56	14
	CDH74	32123	41	87	353	1076	2942	5823	7079	7893	4503	1915	348	63
	CDH80	15398	2	7	42	259	1113	2912	3731	4413	2185	675	56	3
Wind	WSAvg	5.6	6.3	6.7	6.8	6.6	6.0	5.3	4.9	4.9	4.1	4.7	5.2	5.5
Precipitation	PrecAvg	40.1	3.2	2.9	2.9	3.2	4.2	3.8	2.4	2.6	3.7	4.7	3.5	3.1
	PrecMax	56.9	11.3	7.8	6.7	7.0	10.5	11.1	7.0	7.4	8.1	15.2	10.3	10.0
	PrecMin	21.7	0.2	0.2	1.1	0.6	0.6	0.0	0.0	0.4	0.4	0.0	0.4	0.9
	PrecStd	10.5	2.4	2.0	1.4	1.9	2.5	2.6	1.8	1.9	2.2	3.8	2.7	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.7	81.1	83.7	90.2	94.1	100.3	100.8	103.0	100.4	93.3	84.3	79.4
		MCWB	63.2	63.8	67.2	68.2	72.6	72.5	73.1	76.0	73.0	72.4	69.5	65.1
	2%	DB	73.5	76.8	81.2	86.1	91.1	97.2	99.0	100.3	96.9	89.9	81.5	75.2
		MCWB	61.6	63.7	65.1	68.5	72.9	72.5	73.3	74.7	72.7	72.4	68.0	64.1
	5%	DB	70.5	73.0	78.6	82.2	89.6	95.0	96.9	98.8	93.4	86.8	78.7	72.3
		MCWB	61.6	63.2	64.9	68.1	72.3	73.6	73.8	74.5	73.3	71.4	66.5	64.0
	10%	DB	67.8	70.2	74.9	80.7	86.4	92.1	94.7	96.7	90.8	83.9	74.6	69.5
		MCWB	60.5	62.3	64.6	66.9	72.0	73.8	74.3	74.4	72.8	68.9	65.9	62.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.7	69.3	70.8	74.5	78.2	81.0	80.1	80.6	78.9	77.3	73.4	70.8
		MCDB	71.4	73.8	78.0	81.8	87.0	90.2	91.6	87.1	87.2	83.7	79.9	74.5
	2%	WB	66.4	67.6	69.3	72.5	76.5	78.2	78.8	79.5	77.7	75.9	71.4	68.7
		MCDB	70.8	73.1	74.6	80.4	84.9	86.8	88.6	88.4	85.6	83.2	76.7	72.8
	5%	WB	64.7	66.1	68.2	71.3	75.2	77.1	77.9	78.3	76.7	74.5	69.9	66.4
		MCDB	69.1	71.4	73.8	79.2	83.4	86.1	86.8	88.5	84.9	82.0	74.6	70.8
	10%	WB	61.8	64.0	66.8	70.2	73.9	76.2	77.0	77.5	75.7	73.3	68.3	64.0
		MCDB	65.7	68.1	72.6	77.1	82.1	85.0	85.7	88.4	84.4	80.5	73.3	67.3
Mean Daily Temperature Range	5% DB	MDBR	20.6	19.6	20.5	20.7	19.3	19.7	20.1	21.6	20.7	22.2	21.1	20.2
		MCDBR	23.8	22.2	21.4	21.4	20.9	22.7	23.5	24.2	23.7	22.7	22.3	23.5
		MCWBR	14.0	13.3	10.2	9.3	6.4	4.8	4.7	4.6	6.1	8.7	11.7	14.8
	5% WB	MCDBR	18.1	18.4	17.1	17.0	16.8	17.7	18.5	21.5	19.8	18.7	17.9	19.1
		MCWBR	13.8	13.3	10.8	8.6	6.9	5.6	4.6	4.8	5.6	8.9	11.7	15.1
Clear-Sky Solar Irradiance	taub	0.329	0.340	0.364	0.407	0.457	0.437	0.453	0.443	0.420	0.374	0.367	0.323	
	taud	2.489	2.471	2.406	2.301	2.207	2.312	2.302	2.318	2.335	2.447	2.400	2.518	
	Ebn at Noon	287	294	292	281	267	271	266	268	271	278	269	284	
	Edh at Noon	29	32	36	42	46	42	42	41	39	32	31	27	
All-Sky Solar Radiation	RadAvg	931	1061	1405	1694	1835	2049	2068	1955	1631	1357	1013	822	
	RadStd	96	197	128	140	143	143	166	107	151	177	113	104	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+184	+158

Nomenclature: See separate page